

Concept of variation



- 1 Suppose the constant of variation, k , is positive.
 - a If y varies directly with x , describe how y changes as x increases and decreases.
 - b If y varies inversely with x , describe how y changes as x increases and decreases.
- 2 State whether the following equations represent direct or inverse variation:
 - a $y = 5x$
 - b $y = \frac{5}{x}$
- 3 The period of a pendulum varies directly with the square root of its length. If the length is quadrupled, what happens to the period?
- 4 State whether the following are examples of direct or inverse variation:
 - a The variation relating the distance between two locations on a map and the actual distance between the two locations.
 - b The variation relating the number of workers hired to build a house and the time required to build the house.
 - c The variation relating the time it takes an ice cube to melt in water and the temperature of the water.

Direct variation

- 5 In the equation $y = 3x$, y varies directly as x .
 - a Find the value of y when $x = 10$.
 - b Find the value of y when $x = 5$.
 - c For these two ordered pairs, what is the result when the y value is divided by the x value?
- 6 Find the equation relating p and q given the table of values:

p	3	6	9	12
q	$\frac{2}{9}$	$\frac{1}{18}$	$\frac{2}{81}$	$\frac{1}{72}$
- 7 Find the equation relating t and s given the table of values:

s	1	2	3	4
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- 8 If $d = \frac{1}{5}s$, where d is the approximate distance (in miles) from a storm, and s is the number of seconds between seeing lightning and hearing thunder, describe the relationship between the distance and the number of seconds.
- 9 If $r = \frac{d}{t}$, where r is the speed when d kilometres in t hours, describe the relationship between the speed and the kilometres.
- 10 If $f = \frac{mv^2}{r}$, where f is the centripetal force of an object of mass m moving along a circle of radius r at velocity v , describe the relationship between the centripetal force and the mass.
- 11 The mass in grams, M , of a cube of cork varies directly with the cube of the side length in centimetres, x . If a cubic centimetre of cork has a mass of 0.29:
- Find the constant of variation, k .
 - Express M in terms of x .
 - Find the mass of a cube of cork with a side length of 8 centimetres correct to two decimal places.
- 12 The surface area, A , of a regular tetrahedron varies directly with the square of its side length, s . A particular tetrahedron with a side length of 2 cm has a surface area of 6.93.
- Find the constant of variation, k , to two decimal places.
 - Using the rounded value of k , express A in terms of s .
 - Find the surface area of a tetrahedron with a side length of 3 cm to two decimal places.
- 13 The area, A , of an equilateral triangle varies directly with the square of its side length, s . An equilateral triangle with a side length of 7 cm has an area of 21.22.
- Find the constant of variation, k , to two decimal places.
 - Express A in terms of s .
 - Find the area of an equilateral triangle with a side length of 2 cm to two decimal places.
- 14 The number of eggs, n , used in a recipe for a particular cake varies directly with the square of the diameter of the tin, d , for tins with constant depth. 2 eggs are used in a recipe for a tin with a diameter of 17 cm.
- Find the exact value of the constant of variation, k .
 - How many eggs, n , would be used for a tin with a diameter of 39 cm?



Inverse variation

- 15** State the meaning of inverse proportion.
- 16** If r varies inversely with a , write an equation that uses k as the constant of variation.
- 17** In the equation $y = \frac{18}{x}$, y varies inversely with x . When $x = 6$, $y = 3$.
- a** Solve for y when $x = 2$.
 - b** For these two ordered pairs, what is the result when the y -value is multiplied by the x -value?
- 18** Suppose that y varies inversely as the square of x , and that $y = 0.0023$ when $x = 0.4$.
- a** Find the constant of variation, k . Round your answer correct to six decimal places.
 - b** Write the variation equation for y in terms of x .

- 19** State whether the following equations are examples of inverse variation:

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| <p>a $y = \frac{7}{x}$</p> <p>c $y = -\frac{9}{x}$</p> <p>e $y = 2x^2 - 7x - 4$</p> <p>g $x = 1 + y^3$</p> <p>i $y = 6x + 8$</p> <p>k $x = \frac{2}{y}$</p> | <p>b $y = 6x + 8$</p> <p>d $y = \frac{8}{x^2}$</p> <p>f $y = 3 - x$</p> <p>h $x = \frac{8}{y^2}$</p> <p>j $xy = -7$</p> <p>l $xy = 5x$</p> |
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- 20** Consider the inverse variation equation $y = \frac{6}{x}$.

- a** Complete the following tables:

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
y					

- b** Plot the data from the table of values on a number plane.

21 If y is inversely proportional to x , and $y = \frac{375}{t}$ when $x = 10$:

- a Find the constant of variation, k .
- b Express y in terms of x .
- c Find the value of y when $x = 5$.

22 Which of the following tables could represent an inversely proportional relationship between x and y ?

A

x	1	2	3	4
y	3	1.5	1	0.75

B

x	1	2	3	4
y	36	18	12	9

C

x	1	5	6	10
y	3	75	108	300

D

x	1	2	3	4
y	4	5	6	7

23 Find the equation relating n and r given the table of values:

n	1	2	3	4
r	5	$\frac{5}{2}$	$\frac{5}{3}$	$\frac{5}{4}$

24 Consider the equation $s = \frac{375}{t}$.

- a State the constant of proportionality.
- b Find the value of s when $t = 6$.
- c Find the value of s when $t = 12$.

25 The time, t , taken by a typist to type up a document is inversely proportional to his typing speed, s . That is, the quicker the typing speed, the less time it will take. If it takes a typist 20 minutes to type a particular document, typing at a speed of 61 words per minute:

- a Find the constant of variation k .
- b How long (in minutes) will it take a typist with a typing speed of 30.5 words per minute to type up the document?